Suppose now $\vec{E}=0$, $B\neq 0$: $\vec{B}=B\hat{z}$.

Thus $I_2=0$, $I_1=2B^2>0$.

$$\lambda^2 = -\frac{q^2}{4m^2}\left(I_1 \mp I_1\right) = \begin{cases} 0 & (-) \\ -\frac{q^2}{2m^2} I_1 & (+) \end{cases} = \begin{cases} 0 \\ -\frac{q^2B^2}{m^2} \end{cases}.$$

$$\lambda = \pm i\omega \quad , \omega = \frac{qB}{m}$$

$$\frac{d}{d\tau}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \omega & 0 \\ 0 & -\omega & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix}$$

$$\Rightarrow \quad u_0(\tau) = \text{const} \quad , u_z(\tau) = \text{const}.$$

$$\Downarrow$$

$$\frac{d}{d\tau}(\gamma v_z) = 0 \Rightarrow \frac{d}{d\tau}(m\gamma v_z) = \frac{d}{d\tau}p_z = 0.$$

$$u_z(\tau) = u_z(0) = \frac{p_{0z}}{m}$$

$$z(\tau) = \frac{p_{0z}}{m}\tau.$$

Also, $u_0 = \gamma c = \gamma c \frac{mc}{mc} = \frac{\gamma mc^2}{mc} = \frac{\mathcal{E}}{mc}$

$\Rightarrow u_0(\tau) = u_0(0)$ means $\frac{\mathcal{E}(\tau)}{mc} = \frac{\mathcal{E}(0)}{mc} = \frac{\mathcal{E}_0}{mc}$

But $x_0 = ct \Rightarrow u_0 = \frac{dx_0}{d\tau} = c\frac{dt}{d\tau}$

$\Rightarrow$ if $u_0(\tau) = u_0 \Rightarrow ct(\tau) = u_0\tau$

$$t = \frac{u_0}{c}\tau = \frac{\mathcal{E}_0}{mc^2}\tau.$$

Note that $\mathcal{E} = \sqrt{m^2c^4 + c^2p_0^2} > mc^2$

$\Rightarrow \quad t > \tau$ (time in lab frame goes faster)

$$\frac{d}{d\tau}\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \omega \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} u_x \\ u_y \end{pmatrix}$$

$$\lambda = \pm i\omega$$

$\lambda = +i\omega$:
$$\begin{pmatrix} -\lambda & \omega \\ -\omega & -\lambda \end{pmatrix}\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \begin{pmatrix} -i\omega & \omega \\ -\omega & -i\omega \end{pmatrix}\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \omega \begin{pmatrix} -i & 1 \\ -1 & -i \end{pmatrix}\begin{pmatrix} u_x \\ u_y \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} u_x \\ u_y \end{pmatrix} = a \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$\lambda = -i\omega$:
$$\omega \begin{pmatrix} i & 1 \\ -1 & i \end{pmatrix}\begin{pmatrix} u_x \\ u_y \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} u_x \\ u_y \end{pmatrix} = b \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_x(\tau) \\ u_y(\tau) \end{pmatrix} = a \begin{pmatrix} 1 \\ i \end{pmatrix} e^{i\omega\tau} + b \begin{pmatrix} 1 \\ -i \end{pmatrix} e^{-i\omega\tau}$$

Initial condition
$$\begin{pmatrix} u_x(0) \\ u_y(0) \end{pmatrix} = \frac{1}{m}\begin{pmatrix} p_{0x} \\ p_{0y} \end{pmatrix} = \begin{pmatrix} a+b \\ i(a-b) \end{pmatrix}$$

$$\Rightarrow \quad a = \frac{p_{0x} - i p_{0y}}{2m}, \quad b = \frac{p_{0x} + i p_{0y}}{2m}$$

$$u_x(\tau) = \frac{1}{2m}\left((p_{0x} - i p_{0y}) e^{i\omega\tau} + (p_{0x} + i p_{0y}) e^{-i\omega\tau} \right)$$

$$= \frac{p_{0x}}{m} \cos\omega\tau + \frac{p_{0y}}{m} \sin\omega\tau .$$

$$u_y(\tau) = \frac{i}{2m}\left((p_{0x} - i p_{0y}) e^{i\omega\tau} - (p_{0x} + i p_{0y}) e^{-i\omega\tau} \right)$$

$$= \frac{p_{0y}}{m} \cos\omega\tau - \frac{p_{0x}}{m} \sin\omega\tau$$

$$x(\tau) = \int_0^\tau d\tau\, u_x(\tau) = \frac{p_{0x}}{m\omega} \sin\omega\tau + \frac{p_{0y}}{m\omega}(1-\cos\omega\tau)$$

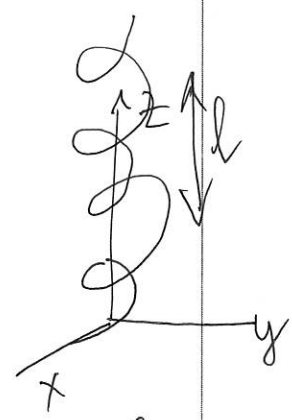
$$y(\tau) = \frac{p_{0y}}{m\omega} \sin\omega\tau + \frac{p_{0x}}{m\omega}(\cos\omega\tau - 1)$$

$$\frac{p_{0x,y}}{m\omega} = \frac{p_{0x,y}}{qB}.$$

Note: $\omega\tau = \omega \frac{mc^2}{\mathcal{E}_0} t = \frac{qB}{m} \frac{mc^2}{\mathcal{E}_0} t = \underbrace{\frac{qBc^2}{\mathcal{E}_0}}_{\omega_0} t$

For the non-relativistic particle $\mathcal{E}_0 \approx mc^2$

$$\Rightarrow \omega_0 = \frac{qBc^2}{\mathcal{E}_0} \Rightarrow \frac{qB}{m}.$$



$$\begin{cases} x(t) = \frac{p_{0x}}{qB} \sin\omega_0 t + \frac{p_{0y}}{qB}(1-\cos\omega_0 t) \\ y(t) = \frac{p_{0y}}{qB} \sin\omega_0 t + \frac{p_{0x}}{qB}(\cos\omega_0 t - 1) \\ z(t) = \frac{p_{0z}}{m} \frac{mc^2}{\mathcal{E}_0} t = \frac{p_{0z}c^2}{\mathcal{E}_0} t \to \frac{p_{0z}}{m} t \text{ for } \mathcal{E}_0 \cong mc^2 \\ t = \frac{\mathcal{E}_0}{mc^2} \tau. \end{cases}$$

Particle moves uniformly along z and spirals around it in (x,y) plane with Larmor frequency ω_0.

Radius of orbit in (x,y) plane: $\frac{\sqrt{p_{0x}^2 + p_{0y}^2}}{qB}$

Step of the spiral along z: $l = \frac{p_{0z}c^2}{\mathcal{E}_0} \cdot \frac{2\pi}{\omega_0} = \frac{p_{0z}c^2}{\mathcal{E}_0} \frac{2\pi}{qBc^2/\mathcal{E}_0} = 2\pi \frac{p_{0z}}{qB}$